

EXERCÍCIOS Em SALA

(5) A PARTIR DO GRÁFICO

MAXIMOS LOCAIS: $(1, 2)$ e $(3, 3)$

MÍNIMO LOCAL: $(2, 1)$

MÁXIMO ABSOLUTO: $(3, 3)$

MÍNIMO ABSOLUTO: $(5, -1)$

(31) $f(x) = 2x^3 - 3x^2 - 36x$

$$f(x) = 6x^2 - 6x - 36$$

$$f(x) = 6(x^2 - x - 6)$$

$$f(x) = 6(x-3)(x+2)$$

NÚMEROS CRÍTICOS:

$$x-3 = 0 \rightarrow 3$$

$$x+2 = 0 \rightarrow -2$$

DERIVADAS

MÍNIMOS E MÁXIMOS

$$C = 1 + f(x) \equiv f(x) = y \cdot x^2$$

Lo y_{MAX} ABSOLUTO

$$C = 6 + f(x) \equiv f(x) = y \cdot x^2$$

VALOR MÍN ABS.

ATIVIDADE EM SALA

33) $y = 1$

$$x^2 = y + 1$$

$$f'(y) = \frac{(y^2 - y + 1)(1) - (1 - 1)(2y - 1)}{y^2 - y + 1}$$

$$\Rightarrow (y^2 - y + 1) - (1 - 1)(2y - 1)$$

$$y^2 - y + 1 - 1 + 1 = y^2 - y + 1$$

$$= y^2 - y + 1$$

$$2y^2 - 3y + 1$$

$$\Rightarrow (y^2 - y + 1) - (2y^2 - 3y + 1)$$

$$y^2 - 1 + 1 - 2y^2 + 3y - 1 = -y^2 + 3y - 1$$

$$-y^2 + 3y - 1$$

$$y^2 - y + 1 + 2y^2 - 3y + 1 = 3y^2 - 4y + 2$$

$$(y^2 - 2y^2) + (1 - y + 3y) + (1 - 1)$$

$$\text{Lo } y^2$$

$$\text{Lo } 2y$$

$$0$$

$$= y^2 + 2y$$

$$y^2 - y + 2 \rightarrow y^2 - y + 2 = 0 \text{ Lo } y^2 + 2y$$

$$b(2) = 10 + 2(1) = 12$$

$$f'(y) = y^2 + 2y$$

$$-1 + 4 = 3$$

$$(y^2 - y + 1)^2$$

11

6/10/11/12/13/14/15/16/17/18/19/20/21/22/23/24/25/26/27/28/29/30/31/32/33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/49/50/51/52/53/54/55/56/57/58/59/60/61/62/63/64/65/66/67/68/69/70/71/72/73/74/75/76/77/78/79/80/81/82/83/84/85/86/87/88/89/90/91/92/93/94/95/96/97/98/99/100

$$f(x) = 3x^4 - 4x^3 - 12x^2 + 7$$

INTERVAL: $[-2, 2]$

1. Deriv.

$$f'(x) = 12x^3 - 12x^2 - 24x$$

$$-4x^3 = +3x^2 = -12x^2$$

$$-12x^2 = -12 \cdot 2x = -24x$$

$$+7 = 0$$

$$f(x) = 12x^3 - 12x^2 - 24x$$

$$\hookrightarrow x^2 - x - 2 = (x-2)(x+1)$$

$$\hookrightarrow 12x(x-2)(x+1)$$

$$\hookrightarrow 12x(x-2)(x+1) = 0$$

$$x = -1, 0, 2$$

$$f(-2) = 3(-2)^4 - 4(-2)^3 - 12(-2)^2 + 7$$

$$= 24 = 16 \rightarrow 3(16) - 4(-8) - 12(4) + 7$$

$$= 2^3 = -8 = 48 + 32 - 48 + 7$$

$$= -2 = 41 = 31$$

$$f(-1) = 3(1) - 4(-1) - 12(1) + 7$$

$$= 3 + 4 - 12 + 7 = -4$$

$$f(0) = 7$$

$$f(2) = 3(16) - 4(8) - 12(4) + 7$$

$$= 48 - 32 - 48 + 7 = -31$$

$$f(3) = 3(81) - 4(27) - 12(9) + 7$$

$$= 243 - 108 - 108 + 7 = 28$$

max min

$$33, -31$$